

P0-A / P0-B / P0-C: step-by-step computational protocols

BACE1_Discovery canvas (print of all tabs)

2026-09-18

Shared BACE1 preparation first, then three independent workflows. Residue numbering is mature BACE1 (Asp32 / Asp228). Run production MD at endosomal pH 4.5 unless a step says otherwise.

Order. Finish Shared + P0-A through MD clustering before P0-B QM/MM or P0-C virtual screening. Do not dock the catalytic dyad as the P0-C pocket.

0. Shared prep

PDB	Why	Use
1W50 (1.75 Å, apo)	Flap-open	Primary receptor
1FKN (OM99-2)	Closed flap + catalytic fingerprint	Anti-target mask; Insert A template
1M4H / P10-P4' StatVal	Long peptide-like ligand	Register reference
3R1G	Exosite Fab	P0-C hotspot #2
2LP1 (NMR C99)	APP TM + juxta	P0-C encounter only
BACE2 (e.g. 2EWY), CatD (1LYB)	Anti-targets	P0-C filter

Prep steps. Chain A ectodomain; mature numbering (Asp32/228; flap 67-77; 10s 9-14; Insert A ~158-167; Insert F ~311-317); rebuild Insert A from 1FKN; ACE/NME; no TMD for P0-A/B; PROPKA3 + PDB2PQR at pH 4.5 then force Asp32-H (OD2) / Asp228; save swapped-dyad replica for 100 ns; ff19SB+TIP3P or CHARMM36m (pick one); ≥12 Å box, 150 mM NaCl; min/heat/NPT as in the canvas (310 K, 1 bar, Langevin, PME 10 Å, 2 fs).

Literature default for the Michaelis complex is Asp32-H / Asp228 (Ellis & Shen 2015 on pH-dependent dyad). If flap RMSD or scissile geometry diverges in the swapped replica, report both.

ID	Sequence (scissile ↓)	Notes
APP_WT_β	SEVKM↓DAEFRHD	APP770 667-678
APP_Swe_β	SEVNL↓DAEFRHD	Swedish; ~60× better — do not overfit
APP_Ice_β	SEVKM↓DTEFRHD	A673T
APP_Path_β	SEVKM↓DVEFRHD	A673V
APP_WT_β'	FRHDSGY↓EVHHQK	Glu11 / C89
SEZ6	YNGRSL↓DVAKAP	Pigoni 2016; 25-mer for AF3
NRG1	KHLGIEF↓MEAEEL	Type I / III-β1 stalk
CHL1	DSIFQ↓DVIETR	Q1061↓D1062
gp130	juxta-TM scan of IL6ST	Site not mapped

AF3/Boltz: 15-25 aa. MD in the cleft: P6-P6' once register is correct. ACE/NME caps.

P0-A — Substrate atlas

Goal: APP-unique contacts and waters. 9 peptide complexes; 3 × 500 ns each; pH 4.5 Asp32-H / Asp228.

Complexes. AF3/Boltz 5 models; accept only productive register (P1 Cα within 6 Å of Asp32 CG; scissile O within 4.5 Å of catalytic water; P1 in S1). HADDOCK/FlexPepDock backup on 1W50 open flap only. Align on BACE1 Cα excluding flap/10s/inserts.

MD. OpenMM 8 or GROMACS 2024; 3 velocities; 500 ns; 310 K; no production restraints (optional 1 kcal on peptide termini then release); coords every 10 ps; do not restrain catalytic water; one 100 ns swapped-dyad APP_WT and SEZ6.

Analysis. Register distances; flap Tyr71-Asp228 CG (closed ~10 Å, open ~15–30 Å); GetContacts occupancy ≥0.4; APP-unique = APP occupancy ≥0.4 and SEZ6 <0.15; MM-GBSA per residue; GIST/SSTMap waters; APBS at pH 4.5.

Done when you export: contact_matrix.csv; app_unique_residues.txt; app_unique_waters.pdb; representative snapshots; β vs β' register plot. If APP and SEZ6 fingerprints match, do not start P0-C.

P0-B — Icelandic mimic

Starting snapshots from P0-A (not 1FKN): WT_β, Ice_β, Path_β, WT_β', SEZ6.

FEP. OpenFE / pmx / FEP+; Ala→Thr, Thr→Val, Ala→Val in complex and free peptide; 12–16 λ; Boresch on peptide backbone vs BACE1 core; 1 ns eq + 5 ns/window; MBAR; overlap <0.03 → add windows. A673T should destabilize β-register or distort P1-P1' vs WT; A673V opposite. If FEP says Ice binds tighter in the β pose, rebuild from β'.

QM/MM. Gem-diol: water attack then cleavage. QM: Asp32, Asp228, catalytic water, P1, P1' (~60–80 atoms). DFT B3LYP-D3(BJ)/def2-SVP opt, def2-TZVP SP. Scan water O-scissile C 2.8 → 1.4 Å. Success: ΔΔE‡ (Ice – WT) at β-site ≈ +2 to +4 kcal/mol; β' barrier for Ice ≤ WT; SEZ6 ΔE‡ unchanged within ~1 kcal/mol.

Ligand design copies Thr, not OM99-2. If the ligand is a hydroxyethylamine in the dyad, you reinvented CTS-21166.

P0-C — PPI / ASBI screen

Pockets from P0-A, not SiteMap on 1FKN alone: (A) APP-unique BACE1 surface with cleft masked; (B) APP peptide / C99 juxta; (C) encounter groove with 2LP1; (D) 3R1G exosite after Fab removal.

MASK: all atoms within 5 Å of Asp32, Asp228, and 1FKN OM99-2.

Ensemble: 5 BACE1 from APP_WT MD + 3 apo 1W50 + 3 APP peptide conformers.

Library: Enamine CNS + ZINC20 in-stock; MW 200–450, cLogP 1–4, TPSA 40–90, HBD ≤2, CNS MPO ≥4, no PAINS, no aspartyl warheads except a tiny FAH focused set. Protonate at pH 5.0 and 7.4.

Docking: GNINA/Vina exhaustiveness 8; keep top 5,000 Bemis–Murcko; hard fail if ligand <3.5 Å from both Asp OD atoms or RMSD <3 Å to 1FKN ligand. Keep ligands that still bind when OM99-2 occupies the dyad and lose score when APP peptide is bound.

Post-dock: top 200 × 3 × 50 ns; top 30 × 200 ns in APP_WT and SEZ6; CatD (1LYB) and BACE2; P-gp/hERG/CYP/hepatotox.

A P0-C hit requires pose in A–D not the dyad, in silico competition, MD-stable 50 ns, APP MM-GBSA better than SEZ6, filters pass, then SPR ± OM99-2 and the enzymatic panel.

Sources

Structures. Patel S, et al. *J Mol Biol.* 2004;343:407–416 (1W50). Hong L, et al. *Science.* 2000;290:150–153 (1FKN). Hong L, et al. *Biochemistry.* 2002;41:10963–10967 (1M4H). Atwal JK, et al. *Sci Transl Med.* 2011;3:84ra43 (3R1G). Barrett PJ, et al. *Science.* 2012;336:1168–1171 (2LP1). CatD 1LYB: Baldwin ET, et al. *Proc Natl Acad Sci USA.* 1993.

Protonation / FF. Olsson MHM, et al. *J Chem Theory Comput.* 2011;7:525–537 (PROPKA3). Jurrus E, et al. *Protein Sci.* 2018;27:112–128 (PDB2PQR). Tian C, et al. *J Chem Theory Comput.* 2020;16:528–552 (ff19SB). Ellis CR, Shen J. *J Am Chem Soc.* 2015;137:9543–9546 (pH-dependent dyad). Default Michaelis: Asp32-H / Asp228.

Peptides. Swedish: Mullan M, et al. *Nat Genet.* 1992. A673T: Jonsson 2012. SEZ6: Pignoni 2016. NRG1: Willem 2006. CHL1: Kuhn 2012.

FEP / QM. Shirts MR, Chodera JD. *J Chem Theory Comput.* 2008 (MBAR). Aspartyl gem-diol path is standard for pepsin-like proteases; BACE1 QM/MM water-attack barriers are typically ~18–22 kcal/mol for WT APP.

Gates. Pratsch 2023 (spines); Zuhl 2016 (CatD); Campagna 2025 (FAH65); Wager 2010 (CNS MPO).